

PART I: NUMBER SYSTEMS

NUMBER-BASE CONVERSIONS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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NUMBER-BASE CONVERSIONS

- Representations of a number in a different radix are said to be equivalent if they have the same decimal representation.
 - Example: $(0011)_8$ and $(1001)_2$ are equivalent. Both have decimal value 9.
- As shown previously, the conversion of a number in base r to decimal is done by expanding the number in a power series and adding all the terms.
- We now present a general procedure for the reverse operation of converting a decimal number to a number in base r .



NUMBER-BASE CONVERSIONS

- Consider a decimal number in the following form:

$$\dots a_2 a_1 a_0 . a_{-1} a_{-2} a_{-3} \dots$$

- The conversion of a decimal integer to a number in base r is done in two parts:
 1. Divide the integer part (left hand side) of the number and all its successive quotients by r and accumulating the remainders until the quotient becomes 0.
 2. Multiply the fraction part (right hand side) of the number with base r , accumulate the integer part of the resultant value and use its fraction part to multiply with base r again. Process continues until either the fraction becomes 0 or the number of digits has sufficient accuracy.



EXAMPLE I: DECIMAL TO BINARY

- Convert $(41.6875)_{10}$ to binary.
- We will split the solution into three parts:
 - The integer part: $(41)_{10}$ to binary.
 - The fraction part: $(0.6875)_{10}$ to binary.
 - Adding the solutions together.

EXAMPLE 1: DECIMAL TO BINARY

I. Convert $(41)_{10}$ to binary.

- 41 is divided by 2 to give an integer quotient of 20 and a remainder of 1.
- Then the quotient is again divided by 2 to give a new quotient and remainder.
- The process is continued until the integer quotient becomes 0.

41		
20	1	a_0
10	0	a_1
5	0	a_2
2	1	a_3
1	0	a_4
0	1	a_5



$$(a_5 a_4 a_3 a_2 a_1 a_0)_2 = (101001)_2$$

EXAMPLE I: DECIMAL TO BINARY

2. Convert $(0.6875)_{10}$ to binary.

	Integer		Fraction	Coefficient
$0.6875 \times 2 =$	1	+	0.3750	$a_{-1} = 1$
$0.3750 \times 2 =$	0	+	0.7500	$a_{-2} = 0$
$0.7500 \times 2 =$	1	+	0.5000	$a_{-3} = 1$
$0.5000 \times 2 =$	1	+	0.0000	$a_{-4} = 1$

$$(0.6875)_{10} = (0.a_{-1}a_{-2}a_{-3}a_{-4})_2 = (0.1011)_2$$

3. Complete solution: $(41.6875)_{10} = (101001.1011)_2$



EXAMPLE 2: DECIMAL TO OCTAL

- Convert $(153)_{10}$ to octal.
 - 153 is divided by 8 to give an integer quotient of 19 and a remainder of 1.
 - Then the quotient is again divided by 8 to give a new quotient and remainder.
 - The process is continued until the integer quotient becomes 0.

153	
19	1
2	3
0	2 = $(231)_8$



EXAMPLE 3: DECIMAL TO OCTAL

- Convert $(0.513)_{10}$ to octal.

$$0.513 \times 8 = 4.104$$

$$0.104 \times 8 = 0.832$$

$$0.832 \times 8 = 6.656$$

$$0.656 \times 8 = 5.248$$

$$0.248 \times 8 = 1.984$$

$$0.984 \times 8 = 7.872$$

$$(0.513)_{10} = (0.406517 \dots)_8$$

ACTIVITY



- Consider your HU ID to be a base 10 number. Convert it directly to hexadecimal and submit it in class.



OCTAL AND HEXADECIMAL NUMBERS

- Binary numbers are difficult to work with because they require three or four times as many digits as their decimal equivalents.
- However, digital computers use binary numbers, and it is sometimes necessary for the human operator or user to communicate directly with the machine by means of such numbers.
- Thus, most computer manuals use either octal or hexadecimal numbers to specify binary quantities .
- The choice between them is arbitrary, although hexadecimal tends to win out, since it can represent a byte with two digits.



OCTAL AND HEXADECIMAL NUMBERS

- Each octal number can be represented by 3 bits. Example: $(63)_8 \rightarrow 110\ 011$
 - Octal became widely used in computing when systems employed 6-bit, 12-bit, 24-bit or 36-bit words.
- Each hexadecimal number can be represented by 4 bits. Example: $8AF \rightarrow 1000\ 1010\ 1111$
 - All modern computing platforms use 16-, 32-, or 64-bit words so hexadecimal is more commonly used in programming languages today.



EXAMPLE 5: BINARY TO OCTAL

- Partition the binary number into groups of three digits each, starting from the binary point and proceeding to the left and to the right.

$$\begin{array}{ccccccc} (10 & 110 & 001 & 101 & 011 & \cdot & 111 & 100 & 000 & 110)_2 & = & (26153.7406)_8 \\ 2 & 6 & 1 & 5 & 3 & & 7 & 4 & 0 & 6 \end{array}$$

- Conversion octal to binary is done by reversing the preceding procedure. Each octal digit is converted to its 3-digit binary equivalent.

$$\begin{array}{ccccccc} (673.124)_8 & = & (110 & 111 & 011 & \cdot & 001 & 010 & 100)_2 \\ & & 6 & 7 & 3 & & 1 & 2 & 4 \end{array}$$



EXAMPLE 6: BINARY TO HEXADECIMAL

- Conversion from binary to hexadecimal is similar to octal, except that the binary number is divided into groups of four digits.

$$\begin{array}{ccccccc} (10 & 1100 & 0110 & 1011 & \cdot & 1111 & 0010)_2 = (2C6B.F2)_{16} \\ 2 & C & 6 & B & & F & 2 \end{array}$$

- Conversion hexadecimal to binary is done by reversing the preceding procedure. Each hexadecimal digit is converted to its 4-digit binary equivalent.

$$\begin{array}{ccccccc} (306.D)_{16} = (0011 & 0000 & 0110 & \cdot & 1101)_2 \\ & 3 & 0 & 6 & & D \end{array}$$

Numbers with Different Bases

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

